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Roll No. .....

HSMC–401

B. TECH. (ECE)

(FOURTH SEMESTER)

MID SEMESTER

EXAMINATION, March, 2024

OPERATION RESEARCH

Time : 1½ Hours

Maximum Marks : 50

Note : (i) All questions are compulsory.

(ii) Answer any *one* sub-questions among (a) and (b) in each main question.

1. (a) Write down the characteristics and scope of operation research. (CO1)

(b) Write down the History and definition of operation research. (CO1)

P. T. O.

2. (a) Calculate the maximal and minimal value of : (CO1)

$$z = 5x + 3y$$

for the following constraints :

$$x + 2y \leq 14$$

$$3x - y \geq 0$$

$$x - y \leq 2$$

by graphical method.

(b) Solve : (CO1)

Max. :

$$z = 7400x_1 + 8320x_2 + 7500x_3$$

Subject to constraints :

$$x_1 + x_2 + x_3 = 200$$

$$10x_1 + 12x_2 + 10x_3 \leq 20,000$$

$$x_1, x_2, x_3 \geq 0.$$

3. (a) Write down the different phases of operation research. (CO2)

(b) (CO2)

Source	Destination A	B	C	Supply
1	2	7	4	5
2	3	3	1	8
3	5	4	7	7
4	1	6	2	14
Demand	7	9	18	

Find solution by least cost method.

4. (a) In a factory, there are six jobs to perform, each of which should go through two machines A and B, in the order A, B. The processing timings (in hours) for the jobs are given here. You are required to determine the sequence for performing the jobs that would minimize the total elapsed time, T. What is the value of T ? (CO2)

Job	J2	J2	J3	J4	J5	J6
M/c A	1	3	8	5	6	3
M/c B	5	6	3	2	2	10

(b) What do you understand by sequencing ? Write down the assumptions used. (CO2)

5. (a) Solve the following assignment problem shown in Table using Hungarian method. The matrix entries are processing time of each man in hours. (CO2)

	I	II	III	IV	V
A	20	15	18	20	25
B	18	20	12	14	15
C	21	23	25	27	25
D	17	18	21	23	20
E	18	18	16	19	20

(b) Solve the transportation problem by VAM : (CO2)

	Destination			Supply
Source	A	B	C	
1	2	2	3	10
2	4	1	2	15
3	1	3	1	40
Demand	20	15	30	